

Rahil Verma

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EDUCATION

Duke University – B.S.E Biomedical Engineering, B.S. Computer Science *Aug 2022 – May 2026*

Coursework: Computer Architecture · Data Structures and Algorithms · Differential Equations · Linear Algebra · Signals and Systems · Deep Learning

Certifications: Supervised Machine Learning (deeplearning.ai) · Advanced Machine Learning Algorithms (deeplearning.ai) · Professional Machine Learning Engineer Path (Google Cloud)

RESEARCH EXPERIENCE

Zaghloul Lab National Institutes of Health (NIH) – Bioengineering Fellow *May 2024 – Present*

- Investigating mechanisms of traveling brain waves in attention and memory through mathematical modeling
- Developed an end-to-end data processing pipeline to extract traveling waves from iEEG signals leveraging high performance computing, reducing delay for acquisition to visualization from weeks to hours
- Applying an autoencoder machine learning model to explain >70% of variance in traveling waves

You Lab – ML Engineer *Jan 2024 – Present*

- Built a genetic algorithm with active learning and ANNs to optimize microbial production of biomaterials

Duke Applied Machine Learning – ML Engineer *Aug 2023 – Present*

- Building semantic search for the Duke course finder using RAG and sentiment analysis

National Institute for Standards and Technologies (NIST) – Research Fellow *May 2023 – August 2023*

- Conducted experiments to optimize the stability of protein-based pharmaceutical formulations
- Utilized a boosted decision tree to identify potentially unstable drug formulations with a success rate of 93%

Center for Global Women's Health Technologies – Data Scientist *Sept 2022 – May 2023*

- Created an [algorithm](#) to identify and label bioluminescent marked tumors
- Resulted in real time processing of images, cutting hundreds of hours of high-cost labeling

ACTIVITIES

Duke Applied Machine Learning Group – Project Manager *Jan 2024 – Present*

- Selected as PM for all data science projects out of 200+ software engineers
- Launched 4 projects in the first 4 months of 2024

CS201 Data Structures and Algorithms (Duke) – Teaching Assistant *Aug 2023 – May 2024*

- Led a discussion section and weekly office hours on fundamental Object-Oriented principles

Duke Enable – Member *Dec 2022 – May 2024*

- Use 3D scanning, CAD, MATLAB, and medical-grade 3D printers to design prosthetics for patients

PROJECTS

PCapro | [GitHub](#) *Apr 2024 - Current*

- Implemented a genetic algorithm to improve principal component analysis in microbial growth optimization

EEGnet | [GitHub](#) *May 2024*

- Constructed a convolutional neural network to identify hand movement from MI-BCI signals

Link | [GitHub](#) *Jan 2023 – Nov 2024*

- Map-based iOS app that allows users to meet with friends in the busy world of college campuses and cities
- Created an MVP using Swift, Firebase, and Google Maps APIs, sold to Glow in November 2023

TECHNICAL SKILLS

Tools: Git · Linux · Google Cloud [VertexAI · BigQuery · Buckets · Dataflow]

Languages: Python · Java · MATLAB · C · JavaScript · Swift

Frameworks and Libraries: TensorFlow · Scikit-learn · Pandas · Apache Beam · Logisim · Langchain